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Grad-505

Final Project Report

Introduction

This project examines particulate nutrient concentrations in the Laurentian Great Lakes using data collected by the United States Environmental Protection Agency Great Lakes National Program Office (GLNPO) between 2006 and 2019. The dataset includes measurements of Particulate Total Nitrogen (PTN), Particulate Total Phosphorus (PTP), Particulate Organic Carbon (POC), Phosphorus Extractable as P (EXP), and Total Suspended Solids (TSS), along with associated spatial and temporal variables such as lake, station, latitude, longitude, sampling depth, season, and year. Because sampling procedures were standardized across lakes and conducted by a single agency and laboratory, the dataset supports meaningful inter-lake comparisons while also allowing for lake-specific analyses.

The overall goal of this project is to understand how particulate nutrient dynamics vary across space, time, and environmental context. Specifically, the analysis addresses three primary research questions. First, I investigate whether nutrient particle concentrations in Lake Michigan have changed over time using linear regression models on log-transformed concentration data.
This question focuses on long-term trends and assesses whether current conditions can reasonably be inferred from historical patterns. Second, I evaluate whether particulate nutrient concentrations combined with seasonal information can accurately predict the lake from which a sample was collected, using classification models and permutation testing to assess predictive performance. Third, I examine whether extreme observations represent atypical environmental conditions and whether season contributes to the likelihood of outlier occurrence after accounting for lake effects.

Exploratory data analysis revealed several important characteristics of the dataset that shaped the modeling approach. Many nutrient variables are right-skewed and contain upper outliers, supporting the use of log transformations and robust summary statistics such as medians. Missingness was also observed in selected variables, requiring careful consideration of sample size and potential bias. These preliminary findings motivated a structured approach that combined regression diagnostics, classification validation using held-out test data, and logistic modeling for anomaly detection.

Overall, this project integrates time-series analysis, supervised classification, and anomaly detection to explore how particulate nutrients behave across the Great Lakes. By combining statistical modeling with environmental context, the analysis aims to distinguish stable long-term patterns from lake-specific differences and seasonally driven anomalies. Through this work, I seek to better understand nutrient dynamics in the Great Lakes and demonstrate how multiple statistical techniques can be applied cohesively to address complex environmental data questions.

Research Questions & Hypotheses

Question 1: How have nutrient particle levels changed over time in Lake Michigan? What predictions, or assumptions, can I make about today's levels based on past trends?

H0: Nutrient particle levels in Lake Michigan are not changing over time. $\beta1 = 0$

H1: Nutrient particle levels in Lake Michigan are changing over time. $\beta1 \neq 0$

Question 2: Can I identify which lake a particulate nutrient sample came from using a combination of particulate nutrient information and seasonal data?

H0: A model using particulate nutrients and season predicts **LAKE no better than a baseline model**.

H1: A model using particulate nutrients and season predicts **LAKE better than the baseline model**.

Question 3: Are there any outliers in the observations that suggest atypical environmental conditions for any of the lakes?
** Atypical environmental conditions could be directly related to the season. Hence the hypothesis. **
H0: After accounting for lake, season does NOT predict outlier status.
H1: After accounting for lake, season DOES predict outlier status.

Analysis with visualizations, tables, and explanatory text

Summary of Analysis – Research Question 1

How have nutrient particle levels changed over time in Lake Michigan, and what predictions or assumptions can be made about current levels based on past trends?
To address this question, I focused specifically on Lake Michigan observations within the GLNPO particulate nutrient dataset. The response variables examined were Particulate Total Nitrogen (PTN), Particulate Total Phosphorus (PTP), Particulate Organic Carbon (POC), and Total Suspended Solids (TSS), selected because they had lower levels of missing data relative to other variables.

Exploratory Data Analysis
Initial exploratory data analysis revealed that all particulate nutrient variables were right-skewed and contained multiple upper outliers (see Figure Q1a). Because skewness and extreme values can distort the assumptions of linear models, log transformations were applied prior to regression.

A preliminary time-series visualization of aggregate median annual TSS values suggested substantial year-to-year variability in Lake Michigan, with no immediately obvious monotonic trend (see Figures Q1b and Q1c). The absence of 2015 data also limited the continuity of the time series, reinforcing the need for caution in interpreting long-term patterns.

Statistical Modeling Approach
To formally test for trends over time, I implemented separate linear regression models for each nutrient, using log-transformed concentrations as the response variable and year as the explanatory variable. The hypotheses were defined as:

- **H_0:** Nutrient particle levels in Lake Michigan are not changing over time ($\beta_1 = 0$).
- **H_1:** Nutrient particle levels in Lake Michigan are changing over time ($\beta_1 \neq 0$).

Before interpreting results, standard regression assumptions were assessed using diagnostic plots, including residual versus fitted plots, Q–Q plots, scale-location plots, and leverage diagnostics. These visuals suggested that linearity, approximate normality, and constant variance were reasonably satisfied for all four nutrients after log transformation (see Figure Q1d).

Because the dataset spans multiple years, independence of residuals was also evaluated using autocorrelation function (ACF) plots (see Figure Q1e). PTN and POC showed some evidence of residual autocorrelation at certain lags, indicating potential temporal dependence. In contrast, PTP and TSS did not display substantial autocorrelation, suggesting that the independence assumption was more clearly satisfied for those models.

Results
Regression results indicated nutrient-specific patterns over time:

- **PTN:** Reject H_0 ($p = 9.558e\text{-}06$). There is strong statistical evidence that PTN levels in Lake Michigan have changed over time. However, residual autocorrelation suggests that the exact p-value should be interpreted with caution.
- **PTP:** Fail to reject H_0 ($p = 0.9541$). No evidence of temporal change.
- **POC:** Fail to reject H_0 ($p = 0.400$). No significant trend detected.
- **TSS:** Fail to reject H_0 ($p = 0.6388$). No evidence of change over time.

Q1 Conclusion
The analysis indicates that three of the four particulate nutrients examined (PTP, POC, and TSS) show no statistically significant long-term trend in Lake Michigan over the study period. This suggests relative stability in phosphorus, organic carbon, and

suspended solids concentrations over time. In contrast, PTN shows strong evidence of temporal change, suggesting that nitrogen dynamics may be evolving differently from those of the other particulate measures.

From a predictive standpoint, these findings suggest that current levels of PTP, POC, and TSS may reasonably be assumed to follow historically stable patterns, while PTN may require closer monitoring due to its detected time-based trend. Overall, Research Question 1 demonstrates that particulate nutrient behavior in Lake Michigan is largely stable across the study period, with the notable exception of nitrogen.

FIGURE Q1a:

EDA Boxplots Confirming Right Skew and Outliers for Each Particulate Nutrient

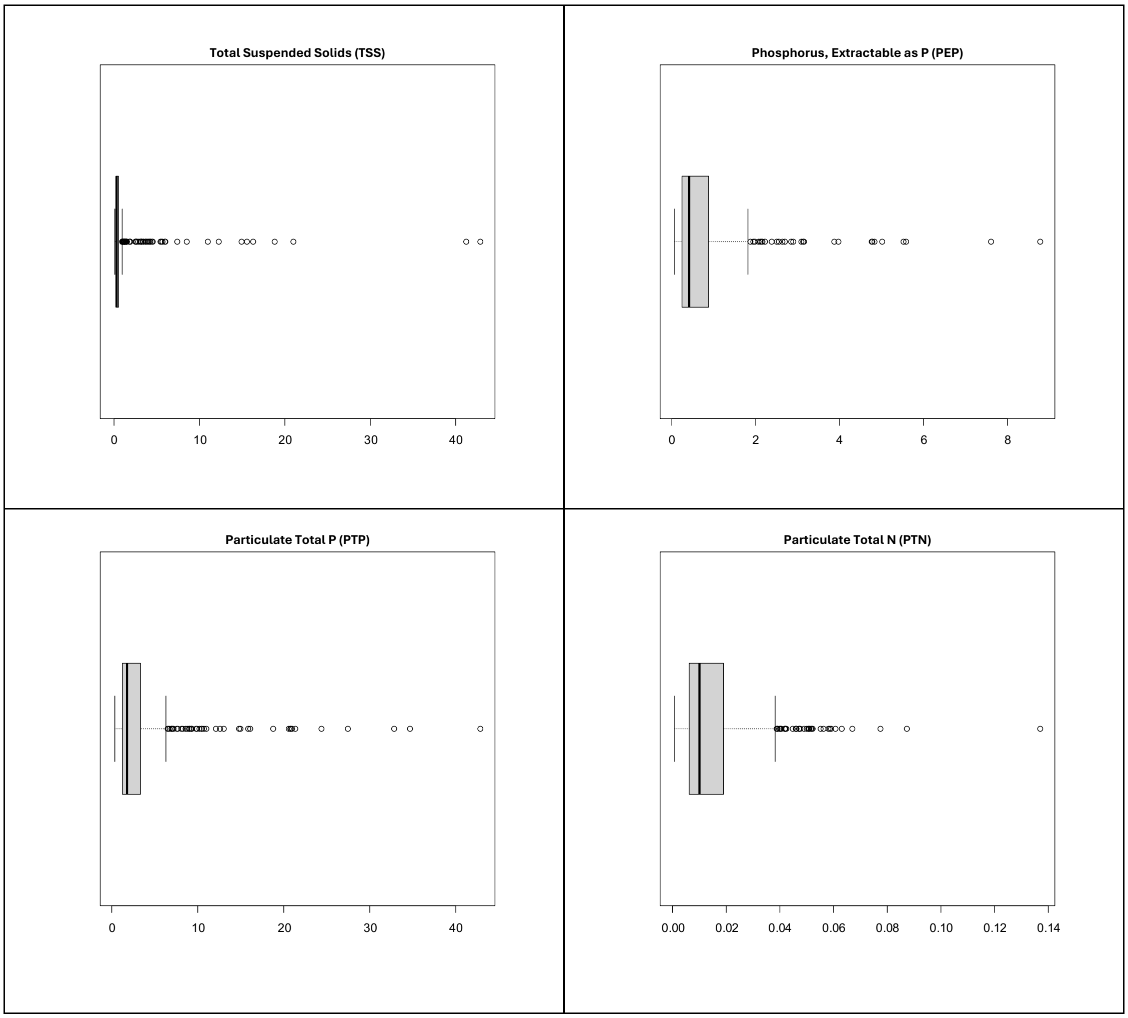


FIGURE Q1b:

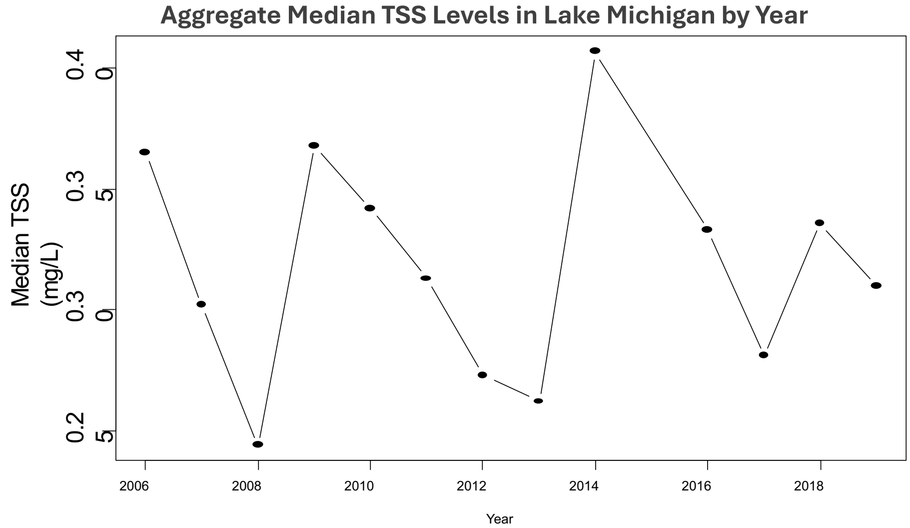


FIGURE Q1c:

Summary Table of Aggregate Meidan TSS Levels by Year

YEAR		MEDIAN	SIZE
01	2006	0.365	9
02	2007	0.302	9
03	2008	0.244	9
04	2009	0.368	9
05	2010	0.342	9
06	2011	0.313	9
07	2012	0.273	9
08	2013	0.262	9
09	2014	0.407	9
10	2016	0.333	9
11	2017	0.281	9
12	2018	0.336	9
13	2019	0.310	9

Figure Q1d:

REGRESSION DIAGNOSTIC PLOTS FOR LAKE MICHIGAN

Each set of four plots checks whether the assumptions for linear regression are reasonable after log transformation. The residuals are mostly centered around zero with no clear patterns, which supports a linear relationship. The Q–Q plots show that residuals are close to normally distributed, with only small deviations at the extremes. The scale–location plots suggest that variability is fairly constant, though PTN and POC show slightly more spread at some fitted values. The residuals vs. leverage plots indicate that no single observation has an unusually large influence on the models. Overall, these diagnostics suggest that the regression assumptions are largely met for all four nutrients.

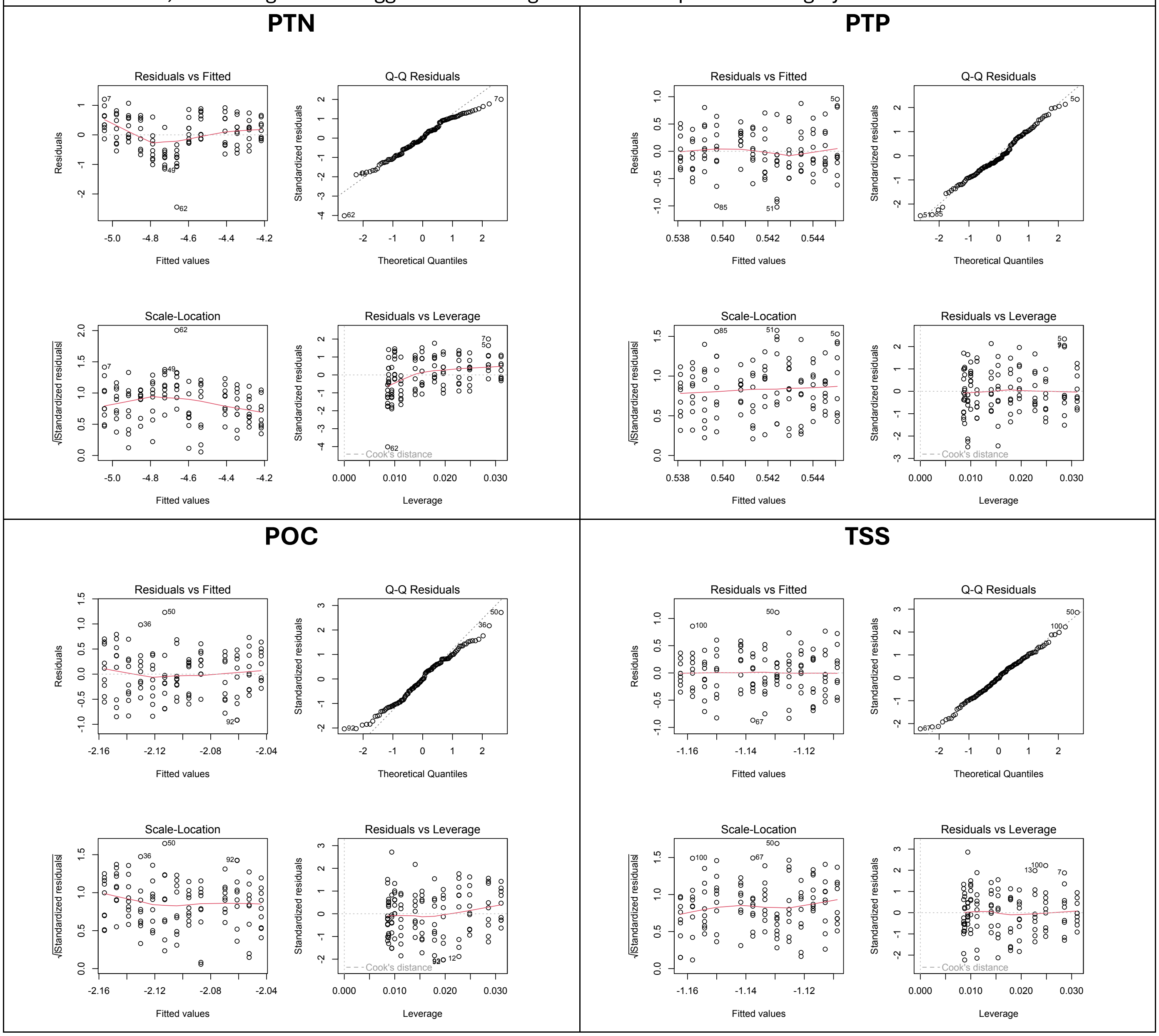
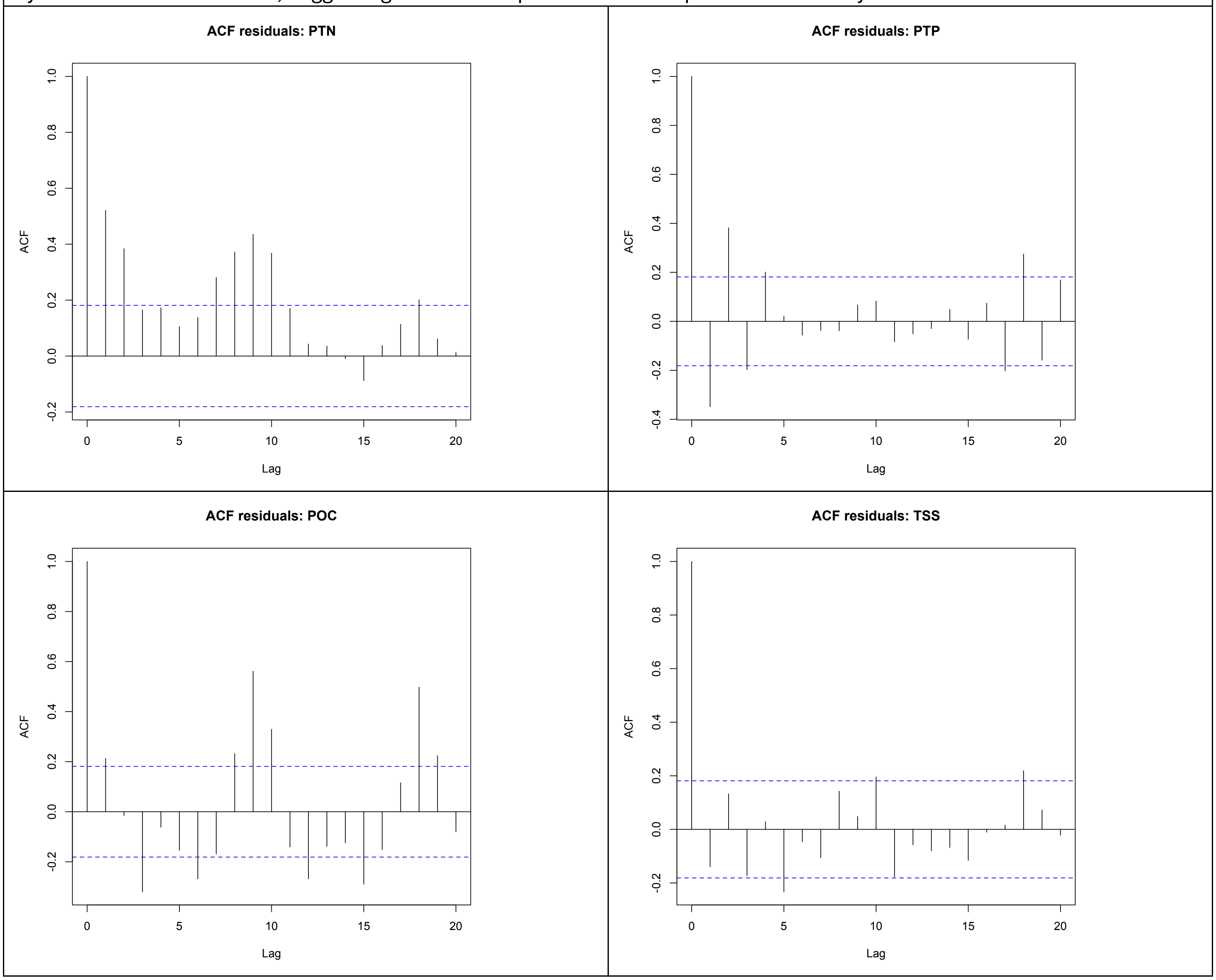


FIGURE Q1e:

ACF OF REGRESSION RESIDUALS FOR PARTICULATE NUTRIENTS IN LAKE MICHIGAN

These autocorrelation function (ACF) plots show residuals from the log-linear regression models for particulate total nitrogen (PTN), particulate total phosphorus (PTP), particulate organic carbon (POC), and total suspended solids (TSS) over time. The dashed blue lines indicate approximate 95% confidence bounds; spikes extending beyond these bounds suggest residual autocorrelation and potential violations of the independence assumption. PTN and POC display several lags with significant autocorrelation, indicating possible temporal dependence in the residuals. In contrast, PTP and TSS show little to no systematic autocorrelation, suggesting that the independence assumption is reasonably satisfied for those models.



Summary of Analysis - Research Question 2:

Can particulate nutrient concentrations, combined with seasonal data, be used to accurately identify which Great Lake a water sample came from?

Exploratory Analysis

Preliminary exploratory data analysis examined differences in a single particulate nutrient concentration across the five Great Lakes. Boxplots of Total Suspended Solids (TSS), displayed on a log scale due to strong right skewness, revealed meaningful differences in medians and interquartile ranges across lakes (see Figures Q2a and Q2b). For example, Lake Erie showed notably higher median values and wider spread compared to several other lakes.

Although distributions overlapped, the observed differences suggested that particulate profiles may contain lake-specific signatures. However, overlap among Lakes Huron, Michigan, and Superior indicated that classification based on a single nutrient would likely be imperfect. These exploratory findings motivated a multivariable classification approach incorporating multiple nutrients and seasonal information.

Statistical Modeling Approach

To formally test predictive capability, I compared three classification models:

1. **Baseline model:** Always predicts the most common lake.
2. **Season-only model:** Uses season (spring vs. summer) as the sole predictor.
3. **Full model:** Uses season plus four log-transformed particulate nutrients (PTN, PTP, POC, and TSS).

The dataset was randomly split into approximately 70% training data and 30% test data to ensure unbiased out-of-sample evaluation. Model performance was assessed using test-set accuracy and confusion matrices.
The hypotheses were defined as:

- **H_0:** A model using particulate nutrients and season predicts lake no better than baseline.
- **H_1:** A model using particulate nutrients and season predicts lake better than baseline.

Because the hypothesis concerns predictive performance rather than mean differences, statistical significance was evaluated using a **permutation test**. Lake labels were randomly shuffled many times (B = 1000), and model accuracy was recalculated for each shuffled dataset to create a null distribution of accuracy under random assignment.

Results

Model accuracies were as follows:

- **Baseline accuracy:** ~0.23
- **Season-only accuracy:** ~0.22
- **Full model accuracy:** ~0.49

The full model more than doubled the baseline accuracy, demonstrating that particulate nutrient measurements contain substantial information about lake identity (see Figure Q2c).

The permutation test yielded a p-value of 0.001, which was the smallest possible value given the number of permutations performed. The observed accuracy exceeded every permuted accuracy value, providing strong evidence that the model captured a real predictive signal rather than random variation. The confusion matrix further clarified classification patterns Lake Erie was classified most accurately, likely due to its distinct particulate profile. In contrast, misclassifications occurred most frequently among Lakes Huron, Michigan, and Superior, suggesting overlapping nutrient distributions among these lakes (see Figure Q2d). Boxplots of log-transformed nutrients by lake and season visually supported this interpretation, showing clear differences alongside substantial overlap (see Figure Q2e / Q3a).

Q2 Conclusion

The analysis provides strong statistical evidence that particulate nutrient concentrations combined with seasonal data can meaningfully predict lake identity. While classification is not perfect due to overlapping distributions among certain lakes, the substantial improvement over baseline demonstrates that each lake exhibits distinguishable particulate characteristics.

In summary, Research Question 2 shows that nutrient dynamics contain measurable lake-specific patterns. These findings reinforce the idea that particulate nutrient profiles reflect underlying environmental and ecological differences across the Great Lakes system.

FIGURE Q2a:

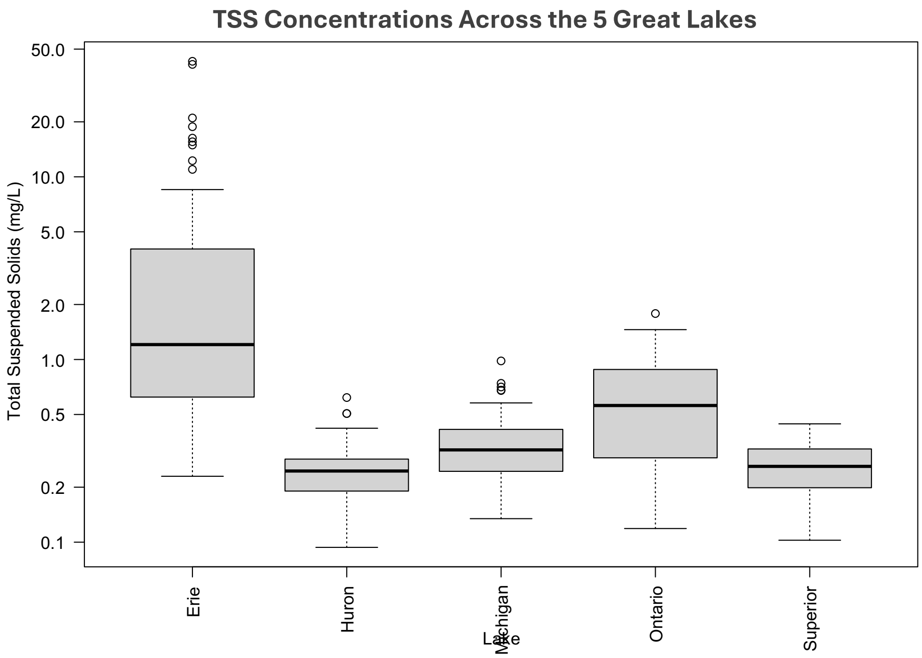


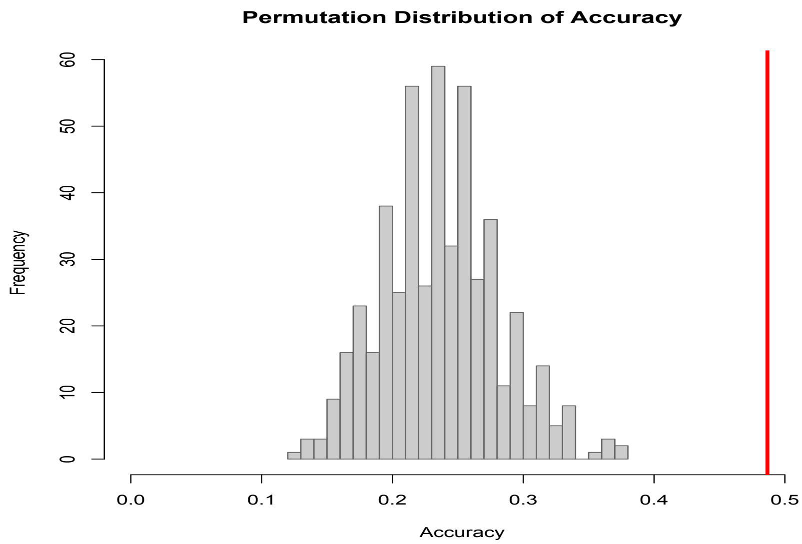
FIGURE Q2b:

Summary Table of TSS Concentration Median and IQR Values by Lake

LAKE	MEDIAN	IQR	SIZE
01 Erie:	1.207	3.404	83
02 Huron:	0.245	0.092	112
03 Michigan:	0.320	0.170	117
04 Ontario:	0.560	0.592	73
05 Superior:	0.260	0.125	115

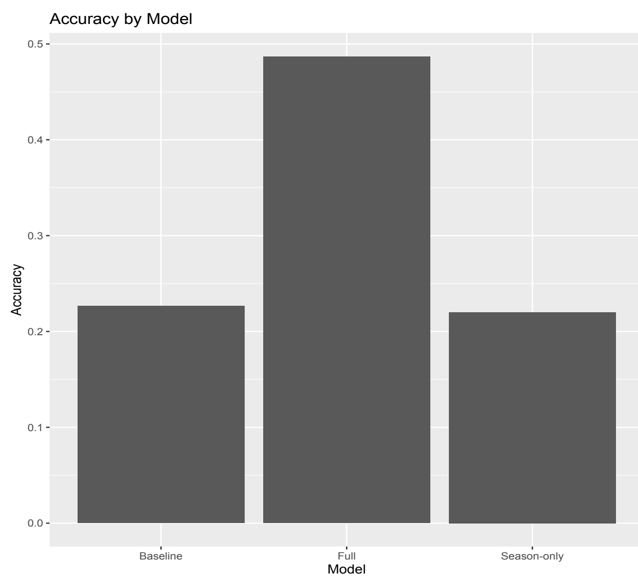
Figure Q2c:

PERMUTATION DISTRIBUTION ACCURACY



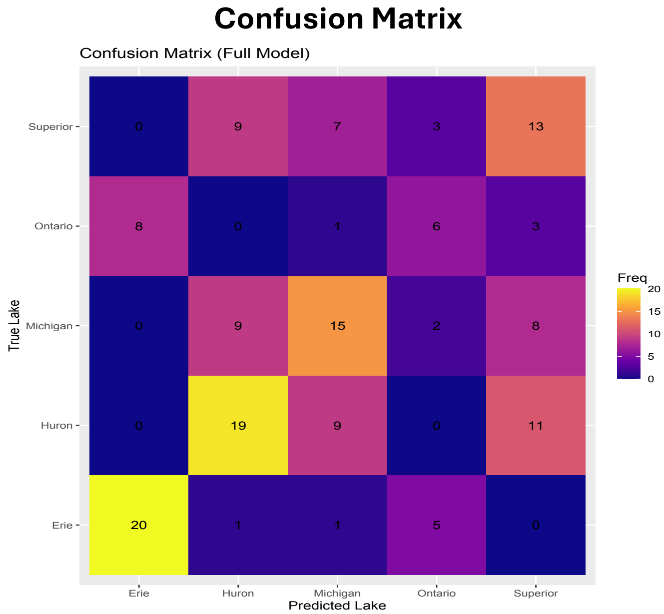
The observed model accuracy of ~48% was outside the range of the permutation distribution, so the x-axis had to be expanded to display it (see the red line).

ACCURACY BY MODEL



Test-set accuracy for the baseline, season-only, and full models. The full model substantially outperforms both the baseline and season-only models, demonstrating that particulate nutrient variables add meaningful predictive information beyond season.

Figure Q2d:



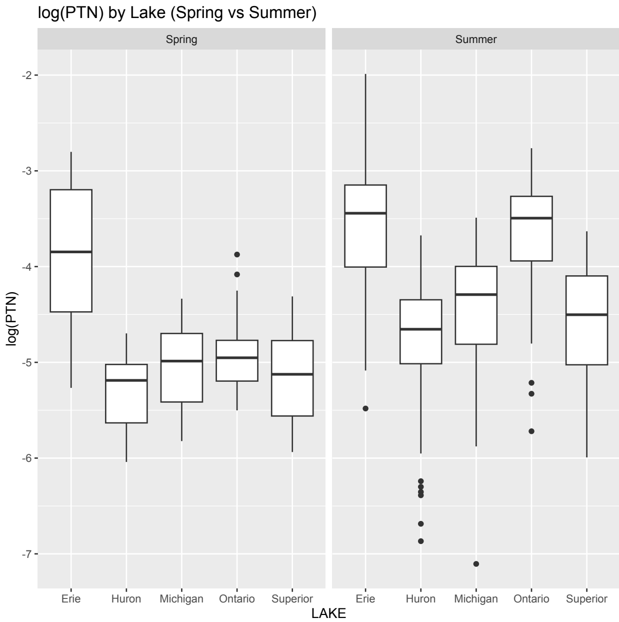
Confusion matrix showing predicted versus true lake labels for the test set (n = 150). Correct classifications appear along the diagonal (upper right to bottom left corners). Lake Erie is classified most accurately, while most misclassifications occur among Lakes Huron, Michigan, and Superior, indicating overlapping particulate nutrient profiles.

Figure Q2e:

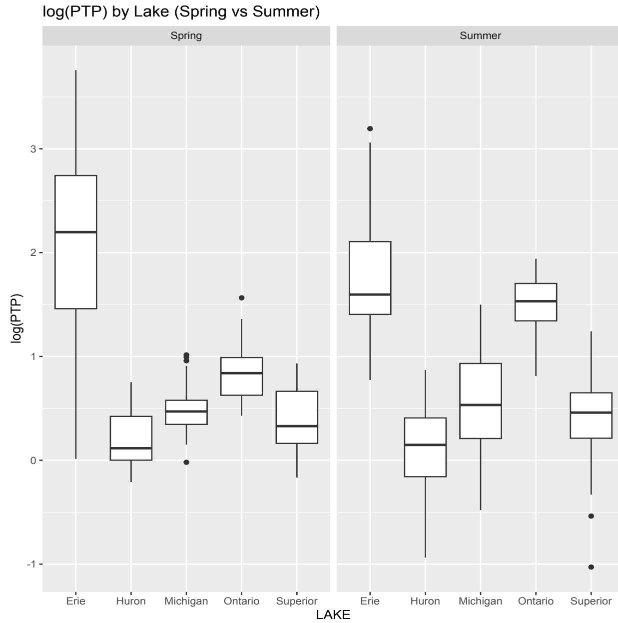
DISTRIBUTION OF LOG() by LAKE (SPRING vs SUMMER) BOXPLOTS

Boxplots of log-transformed particulate nutrients by lake, shown separately for Spring and Summer. Nutrient distributions differ across lakes but overlap substantially, supporting the presence of real signal while explaining imperfect classification performance.

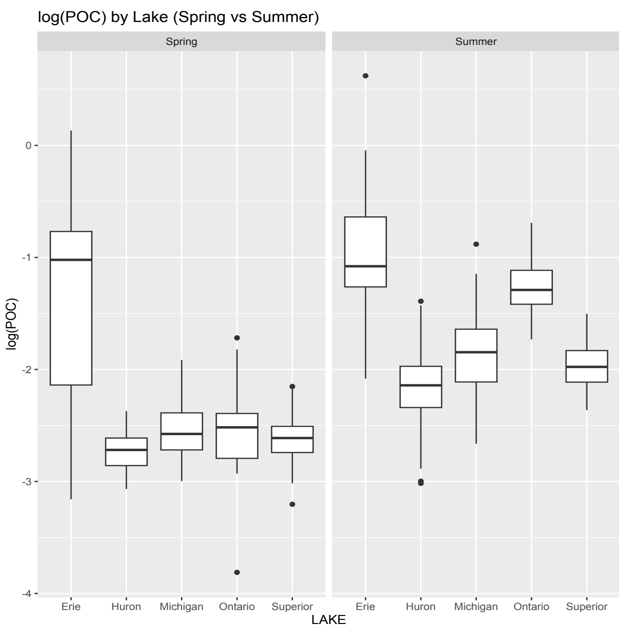
PTN



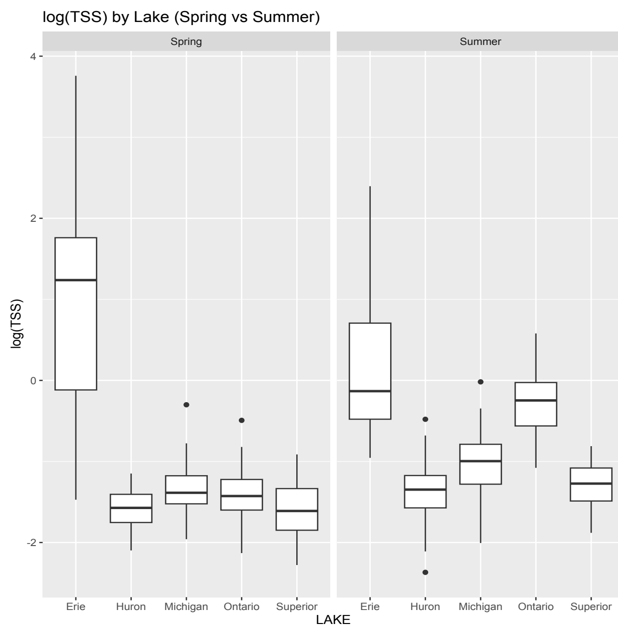
PTP



POC



TSS



Summary of Analysis - Research Question 3:

Are there any outliers in the observations that suggest atypical environmental conditions for any of the lakes, and does season help explain their occurrence?

Exploratory Identification of Outliers

Initial exploratory data analysis revealed that the particulate nutrient variables were right-skewed and contained multiple high-end extreme values. A boxplot by lake showed that all five lakes exhibited upper outliers, although the number and magnitude of these outliers varied across lakes (see Figure Q3a).

Rather than immediately removing extreme observations, the analysis treated them as potentially meaningful environmental events. Outliers for four particulate measures—PTN, PTP, POC, and TSS—were formally defined within each lake using the standard 1.5×IQR rule (see Figure Q3b). Defining outliers separately within each lake ensured that anomalies represented departures from a lake's typical conditions rather than natural differences in baseline concentrations between lakes.

Statistical Modeling Approach

The primary objective was to determine whether season (spring vs. summer) predicts the likelihood of an observation being classified as an outlier, after accounting for lake.
The hypotheses were defined as:

- **H_0:** After accounting for lake, season does not predict outlier status.
- **H_1:** After accounting for lake, season does predict outlier status.

Because outlier status is a binary variable (yes/no), logistic regression was used. For each nutrient, two models were compared:

1. **Lake-only model**
2. **Lake + season model**

Likelihood ratio tests were used to assess whether adding season significantly improved model fit. This modeling approach is appropriate for categorical predictors and does not require normality of the original concentration data.

To distinguish measurement error from plausible environmental events, extreme observations were evaluated using ratio checks (PTN/TSS, PTP/TSS, and POC/TSS). Observations with high TSS and reasonable ratios were retained as likely environmental events, while isolated or inconsistent extremes were flagged for review. Analyses were repeated with and without flagged values to ensure conclusions were not driven by a small number of questionable observations.

Results

Likelihood ratio test results showed nutrient-specific patterns:

- **PTN:** Reject H_0 ($p = 0.014$). Season significantly improves model fit after accounting for lake. PTN outliers are unevenly distributed across seasons.
- **PTP:** Fail to reject H_0 ($p = 0.495$). No evidence that season predicts PTP outlier occurrence.
- **POC:** Fail to reject H_0 ($p = 0.424$). No significant seasonal contribution beyond lake.
- **TSS:** Fail to reject H_0 ($p = 0.396$). Outlier occurrence appears driven primarily by lake rather than season.

Bar plots of seasonal outlier rates showed visible differences in proportions across seasons for certain nutrients, particularly PTN, supporting the statistical findings (see Figure Q3c). Outlier distribution plots further illustrated that some lakes, especially Lake Erie, exhibited more extreme particulate measures, although seasonal effects were not universal across all nutrients (see Figure Q3d).

Q3 Conclusion
The results indicate that seasonal effects on anomalous particulate conditions are nutrient-specific rather than universal. Among the four nutrients examined, only PTN showed a statistically significant association between season and outlier occurrence after controlling for lake. In contrast, PTP, POC, and TSS outliers appear to be more strongly influenced by lake-specific environmental characteristics.

Overall, Research Question 3 demonstrates that extreme particulate events in the Great Lakes are not purely random or measurement errors. Instead, they reflect a combination of lake-specific dynamics and, in the case of nitrogen, meaningful seasonal processes.

FIGURE Q3a:

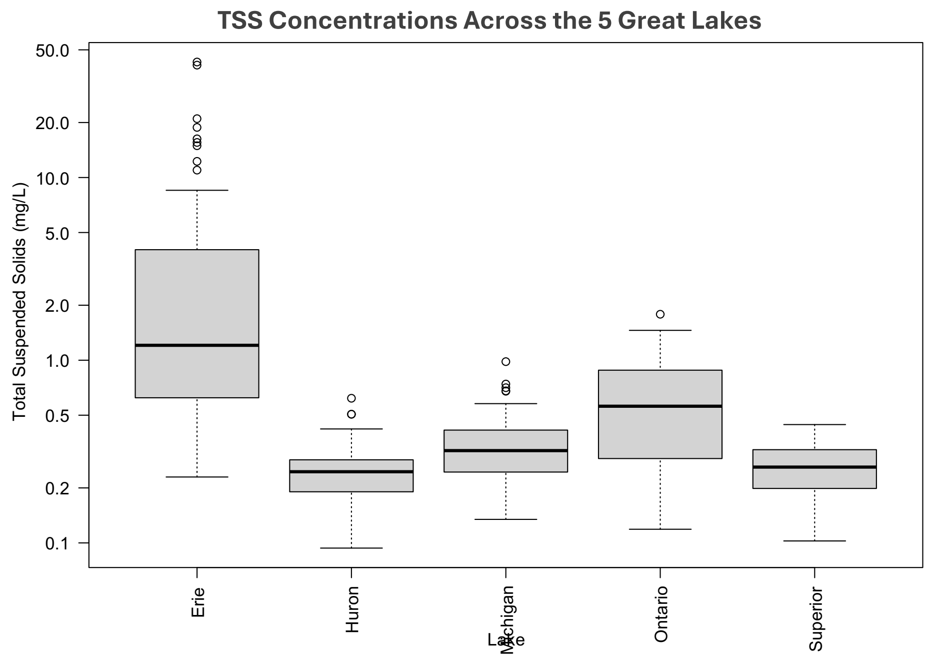


FIGURE 3b:

Summary Table of TSS Concentration Median and IQR Values by Lake

LAKE	MEDIAN	IQR	SIZE
01 Erie:	1.207	3.404	83
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04 Ontario:	0.560	0.592	73
05 Superior:	0.260	0.125	115

FIGURE Q3c:

SEASONAL DIFFERENCES IN OUTLIER RATES FOR PARTICULATE NUTRIENTS

These bar plots show the ***proportion*** of observations flagged as outliers (using the 1.5×IQR rule within each lake) for particulate total nitrogen (PTN), particulate total phosphorus (PTP), particulate organic carbon (POC), and total suspended solids (TSS), separated by season (spring vs. summer). The visuals are used to assess whether outliers occur more frequently in one season, which directly supports the hypothesis test evaluating whether season predicts outlier status after accounting for lake. Overall, some nutrients show higher outlier rates in spring (notably PTP and TSS), while others show higher rates in summer (PTN and POC), suggesting that seasonal processes may contribute to atypical observations rather than outliers being purely random or due to measurement error.

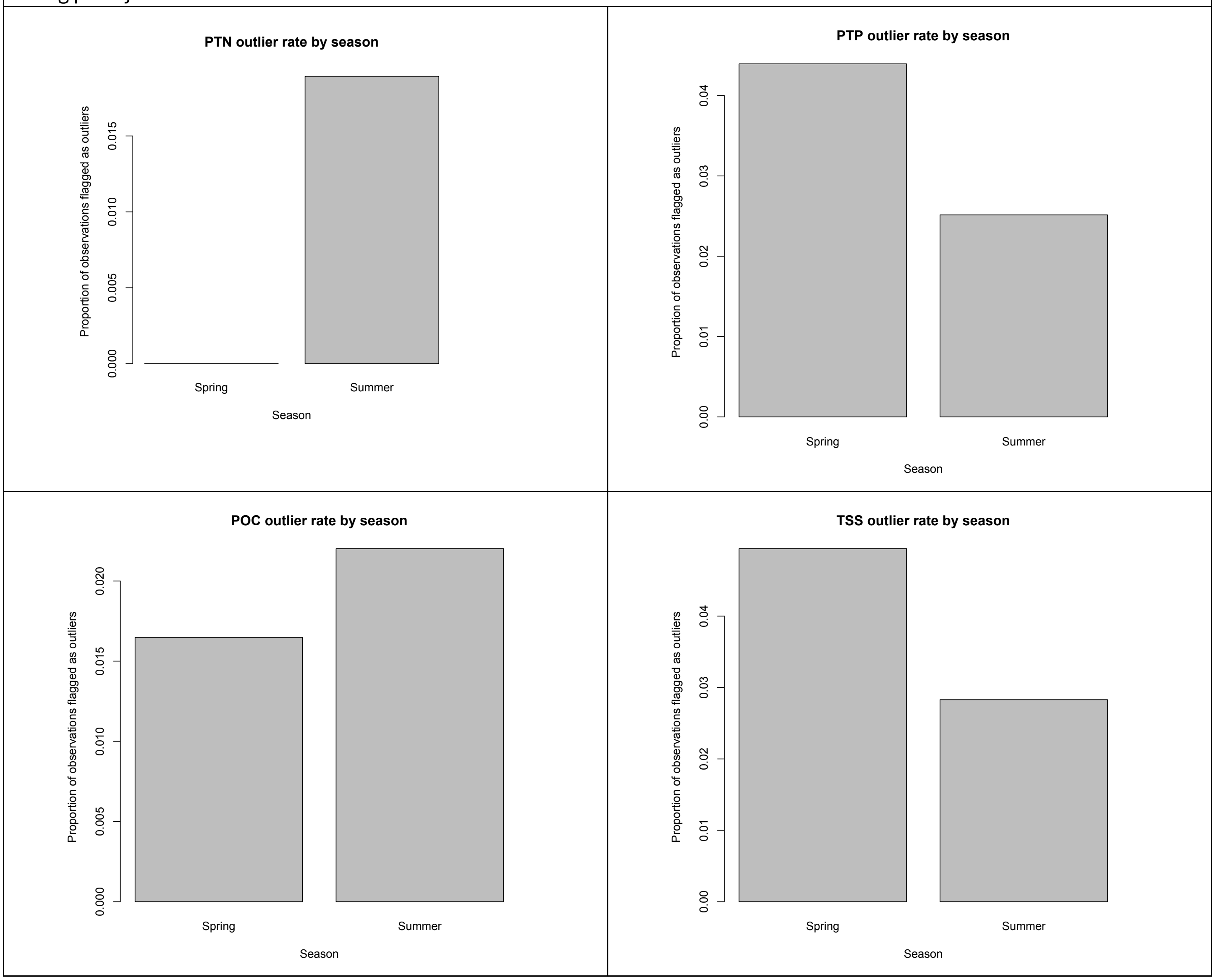

FIGURE Q3d:

OUTLIER DISTRIBUTIONS FOR PARTICULATE NUTRIENTS BY LAKE AND SEASON

Outliers for particulate total nitrogen (PTN), particulate total phosphorus (PTP), particulate organic carbon (POC), and total suspended solids (TSS) were identified using the standard 1.5×IQR rule, calculated separately for each lake. Defining outliers within each lake helps ensure that extreme values reflect unusual conditions relative to that lake, rather than differences in typical nutrient levels across lakes. The points shown represent observations classified as outliers, with color indicating season (spring or summer). The error bars show the interquartile range for each lake–nutrient combination. Together, these plots make it easier to compare how large the extreme values are and whether outliers tend to occur more often in a particular season across different lakes.

*As an aside, I'm really interested in what's driving Lake Erie's extreme particulate measures.